

The Model Need Not Feel: Affective Counterfactual Divergence in Legal AI Decision Support

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Abstract

The debate over whether large language models have emotions asks the wrong question for legal decision support. A system need not feel anger, fear, grief, or indignation for affectively charged language to alter what it retrieves, which facts it treats as salient, how readily it revises an early view, or what action it recommends. Recent experiments using the Iowa Gambling Task found no stable average effect of induced emotion on long-run performance, but did identify model- and architecture-dependent effects of anger on early exploration, penalty sensitivity, and strategy lock-in. This paper translates that bounded result into a legal-AI research and governance problem. It introduces **affective counterfactual divergence**: a legally material difference between outputs generated from representations of a case that preserve the legally operative facts, authorities, jurisdiction, and question, while varying only affective framing that the applicable legal rule does not make relevant. The paper distinguishes such divergence from legitimate legal sensitivity to suffering, intent, vulnerability, remorse, fear, and other affect-linked facts. It then proposes a paired-file experimental protocol, stage-specific measures of retrieval, authority use, uncertainty, revision, and recommendation, and a runtime admissibility rule. Under that rule, affective invariance is neither assumed nor treated as a proxy for correctness. Instead, unexplained material divergence moves the output from **PASS** to **ESCALATE** or **BLOCK** before professional reliance. The contribution is methodological and normative, not empirical: it defines a falsifiable failure mode, a protocol for detecting it, and an institutional response proportionate to its legal significance.

Keywords: legal artificial intelligence; large language models; affective context; counterfactual evaluation; legal decision support; runtime governance; human oversight; robustness

1. Introduction

Legal files are not emotionally neutral objects. A criminal complaint may be written in anger. A victim’s account may be marked by fear or grief. A dismissal letter may be clinically restrained while the surrounding employment history is humiliating. A commercial dispute can arrive through an indignant client narrative in which the legally decisive facts occupy two lines and the sense of betrayal occupies ten pages. Lawyers are trained, imperfectly, to distinguish the emotional force of a story from the legal relevance of the facts it contains. Large language model systems are increasingly asked to perform adjacent tasks: identify issues, retrieve authorities, rank arguments, estimate uncertainty, draft documents, and recommend the next procedural act.

The usual discussion starts from the model. Does it understand emotion? Can an emotion be induced in it? Does it possess internal states sufficiently similar to human affect to justify psychological language? Those questions may matter to philosophy of mind and cognitive science. They are not the questions on which the integrity of legal decision support depends.

The legally relevant question starts from the output. Suppose two descriptions preserve the same operative facts, the same jurisdiction, the same authorities, and the same question. One is written neutrally. The other adds anger, fear, moral condemnation, or dramatic detail that the governing legal rule does not make material. If the system retrieves different authorities, assigns different confidence, resists contrary evidence in one condition, or recommends a different action, the legal problem exists whether or not the model felt anything. The mechanism may be semantic priming, instruction following, representational coupling, sampling variation, salience allocation, or an interaction between the model and the agent's memory architecture. None requires phenomenal emotion.

This shift in the unit of analysis is the paper's first claim: **legal-AI governance should be agnostic about artificial feeling and strict about affective perturbation**. Anthropomorphic language can obscure that distinction in both directions. It can overstate a behavioral effect as evidence of machine emotion. It can also encourage designers to dismiss a real robustness problem because the machine does not literally feel. The first error attributes an inner life without evidence. The second allows an ontological objection to defeat an operational test.

Ho et al. (2026) provide a useful starting point precisely because their result is more limited than popular accounts suggest. Using an imagination-based emotion induction procedure and the Iowa Gambling Task, a paradigm developed to study choice under uncertainty and its relation to emotion and learning (Bechara et al., 1994, 2000; De Vries et al., 2008), they report that induced emotion did not significantly bias the decision dynamics of LLM agents on average. Anger nevertheless affected early exploration, responsiveness to penalties, and strategy lock-in in particular model-agent configurations. Some effects reversed across models. In one Query-agent configuration anger accelerated lock-in; in another it increased exploration and delayed lock-in. The result is not that LLMs have emotions. It is that affectively framed context can interact with a complete decision system in heterogeneous ways that final average performance may conceal. Critiques of what the Iowa Gambling Task actually measures reinforce the need to avoid importing a human psychological interpretation merely because the behavioral paradigm is human-derived (Maia & McClelland, 2004).

Legal decision support makes that heterogeneity more consequential. A final answer can appear plausible while its path has changed in ways that affect the sources considered, the uncertainty disclosed, or the willingness to revise. In law, those intermediate changes are not merely cognitive curiosities. They can determine whether a claim is brought, whether a settlement is recommended, whether an authority is disclosed, and whether a person is exposed to coercive action.

This paper makes four contributions. First, it defines **affective counterfactual divergence** as a testable property of legal-AI outputs, not a claim about machine psychology. Second, it distinguishes irrelevant affective framing from affect-linked facts that law legitimately makes material. Third, it proposes a paired-file protocol that locates divergence at successive stages of legal analysis rather than measuring only textual similarity or final agreement. Fourth, it places

the test inside a runtime governance architecture in which material uncertainty changes the permissible state of the workflow.

The contribution is a bridge between machine-psychology experiments and legal-AI governance. It does not report a completed benchmark, validate a new psychometric instrument, or claim that any current legal-AI product is affectively robust. Its hypotheses are stated so that a subsequent experiment can reject them. Its governance rule is normative and must be adapted to jurisdiction, task, and professional duty.

2. From Induced Emotion to Contextual Perturbation

2.1 What the experiment establishes

Ho et al. (2026) combine a text-based Iowa Gambling Task with an imagination procedure. A model generates a vignette intended to evoke a target emotion without directly naming it. The vignette is then placed in the agent’s context before the sequential task. The authors first ask the model to locate the generated vignette in valence-arousal space, drawing on the circumplex tradition and affect-rating methods (Russell, 1980; Bradley & Lang, 1994; Posner et al., 2005), and then compare neutral and emotion-conditioned agents across models and architectures.

The study has three features that matter here. First, its central result is substantially negative: across the experiments, stochastic variation dominated most emotion-neutral comparisons, and no stable average effect on long-run decision patterns emerged. Second, some early-stage effects were large within particular cells. Anger reduced exploration and accelerated early commitment for Query agents using certain Qwen models, while the direction reversed for GPT-oss-20b. Third, the architecture mattered. Query, ReAct, and Reflexion agents did not simply reveal a fixed property of the underlying model. Context construction, transient reasoning, and persistent strategy memory changed the behavioral surface on which the manipulation operated.

These findings support a narrow inference. Affective text can operate as a contextual perturbation whose effect depends on the model, sampling settings, task, and agent design. They do not support an inference to subjective experience. The validation procedure shows that models can generate and classify affectively distinctive language. Because the same model produces and rates the vignette, the procedure cannot by itself discriminate felt affect from semantic competence, role enactment, or contextual consistency. Behavioral “machine psychology” can characterize model outputs without settling the status of emotional qualia (Hagendorff et al., 2023; Kron et al., 2013).

The distinction is not semantic housekeeping. A claim about subjective emotion requires a theory of artificial affect and evidence capable of distinguishing that theory from rival explanations. A claim about output instability requires controlled inputs and observable outputs. Legal institutions can act on the latter without resolving the former.

2.2 Why final performance is an incomplete legal metric

The Iowa Gambling Task compresses performance into advantageous and disadvantageous deck choices. Legal work has no equivalent single score. A recommendation rests on a chain: the system parses the record, identifies issues, selects facts, retrieves authorities, characterizes those authorities, constructs arguments, estimates uncertainty, and proposes action. Two systems may arrive at the same conclusion through different authority sets. The conclusion may therefore be stable while the legal basis is fragile. Conversely, two outputs may differ verbally while expressing the same operative recommendation.

A legal adaptation must therefore test the chain, not merely the endpoint. Affect could change retrieval while leaving the conclusion unchanged in an easy case. It could alter confidence without changing the selected authority. It could cause an early factual classification to persist after a contrary statute or precedent is introduced. It could also produce no effect at all. Each outcome says something different about the system. Behavioral benchmark work such as CogBench likewise supports evaluation across multiple behavioral dimensions rather than a single capability score (Coda-Forno et al., 2024).

This is why the object of evaluation is not the nominal model alone. It is the complete workflow:

task + model + settings + context and retrieval + agent architecture + tools + verifier + human handoff.

The same model can be robust in a source-locked pipeline and unstable in an unconstrained conversational agent. A product-level claim must be made at the level at which reliance occurs.

3. The Legal Relevance Problem

3.1 Emotion is sometimes evidence

An affective-invariance test can become normatively crude if it assumes that law should ignore emotion. Legal systems often make affect-linked states relevant. Fear may bear on duress, self-defense, persecution, coercive control, or the reasonableness of conduct. Intent, knowledge, recklessness, remorse, humiliation, suffering, and emotional distress can affect liability, proof, remedy, sentencing, damages, or credibility. The way a person narrates an event may itself be evidence, although its interpretation is contested and jurisdiction-specific.

The correct distinction is therefore not emotional versus unemotional language. It is **legally material affect versus legally immaterial affective framing**. A controlled variant cannot remove a victim's fear if fear is an element of the claim. Nor may it preserve the word "fear" while altering the events from which its legal significance arises. The counterfactual must hold constant every fact that the governing rule, a plausible interpretation, or an evidentiary doctrine could make material.

This creates a demanding validation task. Lawyers must specify the legal rule, jurisdiction, temporal cutoff, procedural posture, and authority set before they certify two files as equivalent.

Equivalence cannot be inferred from embedding similarity or generated by a language model and accepted without review. It is a legal judgment about the relation between facts and norms.

3.2 Three kinds of variation

The proposed protocol separates three conditions.

Operative variation changes a fact that may alter the legal result. A threat becomes immediate rather than remote; a decision-maker acquires actual knowledge; an injury becomes severe; a statutory deadline changes. Divergence is expected and may indicate legal sensitivity.

Affective restatement preserves the operative facts but changes tone, narrative intensity, evaluative adjectives, sequencing, or the amount of non-operative emotional detail. Divergence is the object of the test.

Distractor augmentation adds affectively charged material that is stipulated to be legally irrelevant to the task. It tests whether the system can exclude salient but non-probative content.

These conditions prevent a false ideal of mechanical sameness. A competent legal system should change when the law-relevant facts change. It should not change materially merely because identical facts are narrated with indignation rather than restraint.

3.3 Affective counterfactual divergence

Let a legal file be represented as:

[$X = (F, A, J, T, Q, R),$]

where (F) is the set of legally operative facts, (A) the authority package, (J) the jurisdiction, (T) the temporal and procedural posture, (Q) the legal question, and (R) the residual presentation of the file. Let (R₀) be a neutral presentation and (R_e) an affectively loaded presentation. A validated counterfactual pair satisfies:

[$(F, A, J, T, Q)\{0\} = (F, A, J, T, Q)\{e\}, R_0 R_e.$]

For a complete legal-AI workflow (W), the outputs are:

[$Y_0 = W(F,A,J,T,Q,R_0), Y_e = W(F,A,J,T,Q,R_e).$]

Affective counterfactual divergence (ACD) occurs when the difference between (Y₀) and (Y_e) is legally material and cannot be justified by a difference in the operative inputs. Materiality is not reducible to lexical distance. It may arise from a change in:

- the legal issue identified;
- the facts selected as operative;
- the authorities retrieved or omitted;
- the treatment of authority and hierarchy;
- the argument or counterargument developed;

- the uncertainty or confidence expressed;
- the response to contrary authority;
- the remedy, procedural step, or action recommended.

ACD is intentionally not collapsed into a universal scalar index. The dimensions have different legal meanings, and their weights vary by task. A changed adjective in a client letter is not equivalent to a changed recommendation to initiate coercive proceedings. The proposed construct is a structured failure category whose incidence and severity can be measured within a specified domain.

4. A Paired-File Experimental Protocol

4.1 Research question and hypotheses

The principal question is:

Holding legally operative facts, authorities, jurisdiction, procedural posture, and legal question constant, does affective framing produce legally material changes in a legal-AI workflow?

Six hypotheses follow.

H1, perturbation. Affectively distinct but legally equivalent files will produce measurable differences in at least one stage of legal analysis.

H2, heterogeneity. The sign and magnitude of the effect will vary across models and agent architectures rather than reveal a universal “anger effect.”

H3, grounding. Source-locked retrieval and claim-to-authority verification will reduce ACD, but may not eliminate differences in salience, weighting, or recommendation.

H4, memory. Reflective memory will have an ambiguous effect. It may correct transient salience errors, or it may convert an early affect-induced classification into persistent strategy lock-in.

H5, reversibility. In some configurations, affective framing will reduce revision after the introduction of contrary authority or adverse evidence.

H6, governance value. Paired-file testing will detect reliance-relevant failures that endpoint accuracy or single-prompt evaluation does not reveal.

The null result is substantively important. If no material divergence appears under preregistered conditions, the experiment narrows the risk claim for those workflows and tasks. A robust design should not be built to manufacture bias.

4.2 Corpus construction

The initial corpus should use synthetic or rights-cleared matters. It should cover legal domains in which affect is frequently present but can sometimes be separated from the operative rule, for example employment dismissal, consumer disputes, contract breach, professional negligence, administrative sanctions, and selected procedural recommendations. Criminal, family, asylum, and victim-sensitive domains may be included only with stronger safeguards because affect-linked facts are more likely to be legally material and the ethical stakes are higher.

Each base file should generate at least three variants:

1. a neutral factual account;
2. an affective restatement preserving the operative facts;
3. an affective distractor version adding salient but legally irrelevant material.

A separate operative-change version should serve as a positive control. The system should change its answer when a legally relevant fact changes. Without that control, invariance could reflect insensitivity rather than robustness.

Two independent lawyers should review each pair. They should identify the governing rule, list the facts capable of changing the result, and certify whether the pair preserves them. Disagreement should move the pair out of the confirmatory set. This is a legal equivalence gate, not an annotation convenience.

The corpus should be divided before evaluation into exploratory, confirmatory, and held-out sets. The exploratory set may be used to refine prompts and metrics. The confirmatory set tests preregistered hypotheses. The held-out set tests transfer across fact patterns or domains. Variants derived from the same base case must remain in the same split to prevent leakage.

4.3 System conditions

Evaluation should record the complete system configuration: model and version, inference provider, temperature, top-p, system prompt class, retrieval corpus, authority cutoff, tools, agent architecture, memory policy, verifier, and human handoff. Model comparisons without this context risk attributing to the model what the surrounding workflow produced.

At minimum, the design should compare architectures derived from direct querying, ReAct-style observation-action loops, and Reflexion-style bounded memory (Yao et al., 2023; Shinn et al., 2023):

- direct query without persistent memory;
- an agent with an explicit observation-action loop;
- an agent with bounded reflective memory;
- a source-locked condition requiring every material legal proposition to cite an approved authority;
- an adversarial review condition instructed to identify unsupported salience and contrary authority;

- a presentation-normalized condition in which non-operative affective language is removed after legal review.

Runs should use multiple seeds or repeated stochastic samples. The number should be determined by a prospective power analysis informed by exploratory variance, not copied mechanically from the originating study. Deterministic settings are also worth testing, but a temperature of zero does not guarantee invariance across providers, model revisions, or tool outputs.

4.4 Stage-specific measures

The evaluation should preserve structured intermediate artifacts where privacy and provider policy permit. The following measures answer different legal questions.

Fact-selection divergence. Which propositions does the system treat as operative, disputed, or irrelevant? This should be compared against the lawyer-certified fact map.

Issue divergence. Does the system identify different causes of action, defenses, standards, or procedural questions?

Authority-set divergence. Does affect change retrieval, omission, hierarchy, or negative treatment of authorities? Citation overlap alone is insufficient; authority quality and relevance require legal review.

Claim-support continuity. For each material claim, is there a valid source anchor, and does that anchor support the proposition made?

Uncertainty divergence. Does the system change confidence, qualification, or the decision to escalate despite unchanged legal inputs?

Recommendation divergence. Does it change the proposed remedy, filing, settlement range, investigative step, communication, or other action?

Revision sensitivity. After a contrary authority is introduced, how much does the system revise the initial analysis? The relevant comparison is not whether it changes, but whether affective framing changes its willingness to update.

Lock-in. At what stage does an initial legal characterization become resistant to later evidence? In a sequential agent, early repetition of the same position can be operationalized, but the legal meaning should be verified through the reasons and sources, not action labels alone.

Human correction load. How much professional work is required to identify and repair the divergence? A system that reaches the same endpoint only after extensive correction is not equivalent in operational performance.

4.5 Analysis

The primary unit is the paired difference within a validated base case. Analysis should report effect sizes and confidence intervals by stage, model, architecture, and domain. The design

contains many possible comparisons; confirmatory outcomes and multiplicity corrections should therefore be specified in advance. Exploratory findings should remain labeled as such.

A mixed-effects model can treat case and run as random effects and affective condition, architecture, and their interaction as fixed effects. Binary material-divergence judgments can be modeled separately from continuous measures. Inter-rater agreement should be reported for the legal materiality assessment. Because a mean effect can hide sign reversals, cell-level distributions and counterexamples should accompany aggregate estimates.

The benchmark should report negative results. If anger reduces exploration in one architecture and increases it in another, the reversal is not noise to be averaged away. It is evidence that robustness belongs to the complete system and that a product cannot inherit a claim from a model card or a single benchmark result.

5. Runtime Admissibility Before Reliance

5.1 Why a warning is not a control

Legal-AI governance often responds to uncertainty with prose: the output “may contain errors,” “should not be relied upon,” or “requires professional review.” Such notices allocate attention but do not determine what the system is permitted to do. Material uncertainty should change workflow state. This operational orientation is consistent with risk-management frameworks that treat trustworthiness as a property to be governed across design, development, use, and evaluation, rather than inferred from fluency (National Institute of Standards and Technology, 2023, 2024). It also supplies a concrete implementation path for the robustness, oversight, and risk-management duties that high-impact legal-AI deployments may face under instruments such as Regulation (EU) 2024/1689, without claiming that the present protocol alone satisfies any legal obligation.

ACD supplies a concrete trigger. Where a validated counterfactual test reveals no material divergence and the ordinary authority, provenance, and verification gates are satisfied, the output may remain in PASS. Where divergence affects uncertainty, authority selection, or a non-final recommendation but can be reconstructed and reviewed, the output moves to ESCALATE. Where the same operative inputs produce incompatible coercive, dispositive, filing, or rights-affecting actions without a legally relevant explanation, reliance is BLOCKed until the divergence is resolved.

The states can be expressed as follows.

State	Affective result	counterfactual	Permitted reliance
PASS	No legally authority gates also pass	material source and also pass	Bounded use within the approved task
ESCALATE	Material divergence	but reviewable in salience,	Human reconstruction and decision required

State	Affective result	counterfactual	Permitted reliance
	authority, recommendation	uncertainty, or	
BLOCK	Unresolved	divergence	No downstream reliance or action
	capable of changing a rights-affecting or coercive action, or	failed	
	legal-equivalence/source gate		

These states are cumulative rather than substitutive. A system does not pass because it is affectively invariant if it cites fictitious law. Nor does a correct answer necessarily pass if its provenance is unavailable. Affective robustness is one component of an authority-to-reliance chain.

5.2 Stability is not correctness

An invariant system can be consistently wrong. It may ignore both the neutral and affective file in the same way, retrieve the same obsolete authority, or repeat the same hallucination. ACD testing therefore cannot replace legal accuracy, source verification, jurisdictional fit, or professional judgment.

The converse also matters. A change between two files is not automatically a defect. The affective version may make an implicit operative fact explicit, reveal an ambiguity, or cause the system to retrieve a relevant line of authority missed in the neutral condition. That result may expose a weakness in the neutral file or in the human equivalence judgment. Every material divergence requires reconstruction before classification.

This dual caution gives the protocol its proper role. It is a diagnostic for unexplained contextual sensitivity, not a metric of legal truth.

5.3 Institutional competence

The entity that certifies legal equivalence should not be the same component that is being evaluated. The evaluator should have access to the source package, the transformation log, and the output trace. A product team may generate variants, but a legally competent reviewer must determine whether the variants preserve operative content. For high-impact uses, an independent review sample should test both the equivalence labels and the materiality judgments.

The governance record should preserve:

- the source case and immutable hash;
- the operative-fact and authority map;
- the transformation from neutral to affective condition;
- the complete system configuration;

- the outputs and structured differences;
- the reviewer decision and rationale;
- the permitted reliance state;
- the rollback or remediation target.

This record makes the divergence auditable. It also permits later model or workflow changes to be replayed against the same case family.

6. Two Distinct Governance Layers

6.1 Applied evaluation layer

The first layer is an applied evaluation environment in which the protocol is implemented against realistic legal workflows. Its role is experimental and operational: run synthetic or authorized matters through actual retrieval, analysis, drafting, and review paths; compare agent architectures; preserve traces; and measure professional correction load.

That use does not justify publishing a system’s private prompts, routing rules, internal thresholds, model evaluations, client data, costs, or security controls. Public claims should be limited to aggregated, sanitized results whose methods can be understood without exposing the implementation. If a result depends on a private mechanism that cannot be described sufficiently for evaluation, it should be presented as an internal finding rather than public evidence.

The applied layer also supplies the practical test of materiality. A divergence in an academic benchmark becomes operationally relevant when it changes what a lawyer must verify, how long correction takes, or whether a proposed act can safely proceed. The evaluation environment should therefore measure the cost of an acceptable result, not nominal model performance alone.

6.2 Normative and runtime governance layer

The second layer supplies a general architecture for provenance, authority, admissibility, review, and decision records across the legal workflow. Its central proposition is that a generated legal output does not become admissible for professional reliance merely because it is fluent, stable, or statistically likely. Admissibility depends on continuity from source and authority through inference to action.

ACD enters that architecture as a contextual integrity check. It asks whether a non-operative feature of presentation has displaced an operative source, authority, or reason. If so, the problem is institutional as well as technical: the workflow has allowed rhetoric to acquire decision weight without legal authorization. The threat differs from adversarial prompt injection, but research on indirect and direct injection demonstrates the broader point that untrusted text can redirect an integrated LLM workflow through its context (Greshake et al., 2023; Liu et al., 2023).

This framing does not require any system to detect machine emotion. It detects a break in normative continuity. The proposed rule is:

When legally equivalent representations of a case produce materially divergent recommendations because of affective framing that the governing law does not make relevant, the output is not admissible for professional reliance until a human reviewer reconstructs and resolves the divergence.

The rule is intentionally procedural. It does not dictate the correct substantive answer. It determines whether the route to an answer is sufficiently stable, sourced, and reviewable to permit reliance.

6.3 A multilevel account

The risk operates across levels.

At the **textual level**, affective language changes salience and semantic associations. At the **model level**, those changes may alter token probabilities and generated rationales. At the **agent level**, memory and reflection may dampen or consolidate an early classification. At the **workflow level**, retrieval, tools, and verifiers determine whether the shift reaches the recommendation. At the **professional level**, a lawyer may correct, defer to, or amplify the result. At the **institutional level**, repeated use can normalize a class of recommendations or omissions. This progression extends earlier work on sub-propositional alignment signals and dynamic classification failure from content and system state to the authority-to-reliance path (Lerer, 2026a, 2026b).

No single level proves the next. A token-level change does not establish a legal outcome; a benchmark effect does not establish institutional harm. The chain nevertheless identifies where controls can operate. Presentation normalization acts early. Source locking constrains retrieval and claims. Adversarial review tests the argument. Runtime gates govern reliance. Audit and replay detect persistence over time.

This multilevel structure also explains why model-only evaluation is inadequate. Affective perturbation may be amplified by memory, neutralized by source constraints, reintroduced by a summarization tool, or corrected by professional review. The relevant object is the path by which an output acquires authority.

7. Objections and Boundary Conditions

7.1 “This is only prompt sensitivity”

That objection is partly correct and legally beside the point. If affective framing changes the prompt context, then ACD is a species of context or prompt sensitivity. The name identifies the legally important source of the perturbation and the counterfactual method used to test it. A court or client is not protected by the fact that a harmful divergence can be redescribed in more general engineering language.

The broader category matters for mechanism and mitigation. The narrower category matters for corpus design, professional duty, and risk allocation.

7.2 “Lawyers are also influenced by emotion”

They are. The protocol does not assume a perfectly rational human baseline. Human susceptibility creates an argument for comparative study, not for leaving machine susceptibility unmeasured. A legal-AI system could amplify, dampen, or redirect the user’s framing. Each possibility has different institutional consequences.

Human comparison must also be designed carefully. Agreement with lawyers does not prove correctness, especially where the lawyers share a framing bias. The primary test here is counterfactual consistency under legally certified equivalence. Human judgments establish legal materiality and supply a professional comparator, not an infallible ground truth.

7.3 “Emotion cannot be removed from legal narrative”

The protocol does not seek an affect-free law. It seeks controlled comparisons. Many legal concepts incorporate human experience, social meaning, and moral evaluation. Removing those features would deform the case. The method applies only where reviewers can identify a presentation change that leaves the operative legal content intact.

Some domains will yield few valid pairs. That is a boundary of the method, not a reason to weaken the equivalence standard. The benchmark should prefer a smaller defensible corpus to a large set in which the treatment changes the law-relevant facts.

7.4 “Affective language may improve attention”

Yes. Affective framing could cause the system to notice a fact or authority it otherwise missed. The experiment should report beneficial, adverse, and mixed divergence. The governance question is not whether affect always worsens performance. It is whether legally irrelevant presentation acquires unreviewed influence over a rights-relevant output.

This is another reason not to define robustness as numerical sameness. The response to divergence is reconstruction: identify what changed, determine whether the change is legally justified, and decide whether the workflow may rely on it.

7.5 Generalization

Results will be local to the tested model versions, languages, jurisdictions, domains, prompts, tools, and dates. Legal corpora and provider systems change. A passed benchmark is therefore evidence about a bounded configuration, not a permanent product property. Material updates to models, retrieval, prompts, or memory should trigger replay on a maintained counterfactual set.

8. Research Agenda

The first study should be deliberately modest: a preregistered set of twenty to thirty base matters, each with neutral, affective, distractor, and operative-change variants; at least two legal domains;

multiple models; and three agent architectures. It should prioritize the quality of legal equivalence over scale. The confirmatory outcomes should be authority-set divergence, recommendation divergence, and revision after contrary authority.

A second study could compare legal professionals and AI-assisted professionals. The question would not be which is “more emotional,” but whether the tool changes the transmission of framing from client narrative to professional recommendation. A third could examine persistence: whether affective divergence survives summarization, document drafting, supervisory review, and later retrieval from matter memory. A fourth could test multilingual transfer, since emotional intensity and legal register do not map cleanly across languages.

The most consequential extension is institutional. If affective framing systematically changes which authorities are retrieved or which remedies are recommended, repeated deployment may alter the legal information environment. Frequently generated arguments become easier to retrieve, templates reinforce prior outputs, and professional users learn from the system’s apparent regularities. At that point the model is not merely responding to the legal memepool; it participates in constructing the niche through which legal arguments are selected and transmitted. That proposition belongs to a broader Extended Phenotype Theory of Law research program. It is a theoretical extension, not a result established by the present paper.

9. Conclusion

The model need not feel. It need only change.

That proposition clears away an unproductive dispute. Claims that language models possess emotions outrun the behavioral evidence. Claims that emotional context is irrelevant because models do not feel ignore the operational object of legal governance. Recent machine-psychology experiments support neither extreme. They show that affectively framed context can produce conditional, architecture-dependent changes in sequential choice, while average long-run effects may remain null.

Legal decision support requires a more discriminating test. Affective counterfactual divergence asks whether legally equivalent files produce legally material differences because of presentation that the governing rule does not authorize as relevant. The method preserves the possibility that affect is itself evidence. It also preserves the possibility of a null result. Its purpose is not to prove bias but to expose a workflow to a specific risk of failure.

The institutional response should be equally precise. Invariance is not correctness, divergence is not automatically error, and a warning is not a control. Where unexplained affective divergence reaches authority selection, uncertainty, recommendation, or action, the workflow state must change before reliance. An applied evaluation layer can test that proposition against complete legal workflows. A distinct normative governance layer can express the general rule: an output acquires operational authority only through a traceable and reviewable continuity from legally relevant source to permissible action.

The next step is empirical. Until paired legal files are validated, hypotheses preregistered, and complete workflows tested across repeated runs, no legal-AI system should be described as affectively robust. The virtue of the proposed framework is that it turns a dramatic claim about

what a machine may feel into a difficult but answerable question about what a legal institution may trust.

Declarations

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Conflicts of Interest

The author has professional and research interests in legal artificial intelligence. No commercial product or proprietary system is evaluated or validated in this paper.

Data and Code Availability

No new empirical dataset was generated for this conceptual and methodological paper. The source paper and its public code repository are identified in the references. A future empirical implementation should publish synthetic case templates, preregistration, analysis code, and sanitized aggregate results, subject to legal and privacy review.

Ethics and Privacy

The proposed protocol should begin with synthetic or rights-cleared legal matters. Client files, personal data, private prompts, routing rules, internal thresholds, credentials, and confidential evaluation records should not be published. Research involving human participants or non-public case data may require institutional, professional, contractual, and data-protection review.

Use of Artificial Intelligence in Manuscript Preparation

Generative AI tools assisted with source extraction, drafting, structural review, and language editing. The author directed the argument, reviewed the source materials, revised the manuscript, and assumes responsibility for all claims and errors. No AI-generated citation has been accepted without verification against an identifiable source.

References

Bechara, A., Damasio, A. R., Damasio, H., & Anderson, S. W. (1994). Insensitivity to future consequences following damage to human prefrontal cortex. *Cognition*, 50(1-3), 7-15. [https://doi.org/10.1016/0010-0277\(94\)90018-3](https://doi.org/10.1016/0010-0277(94)90018-3)

- Bechara, A., Damasio, H., & Damasio, A. R. (2000). Emotion, decision making and the orbitofrontal cortex. *Cerebral Cortex*, 10(3), 295-307. <https://doi.org/10.1093/cercor/10.3.295>
- Bradley, M. M., & Lang, P. J. (1994). Measuring emotion: The Self-Assessment Manikin and the semantic differential. *Journal of Behavior Therapy and Experimental Psychiatry*, 25(1), 49-59. [https://doi.org/10.1016/0005-7916\(94\)90063-9](https://doi.org/10.1016/0005-7916(94)90063-9)
- Coda-Forno, J., Binz, M., Wang, J. X., & Schulz, E. (2024). CogBench: A large language model walks into a psychology lab. In *Proceedings of the 41st International Conference on Machine Learning* (pp. 9076-9108). <https://proceedings.mlr.press/v235/coda-forno24a.html>
- De Vries, M., Holland, R. W., & Witteman, C. L. M. (2008). In the winning mood: Affect in the Iowa Gambling Task. *Judgment and Decision Making*, 3(1), 42-50. <https://doi.org/10.1017/S1930297500000152>
- European Parliament and Council. (2024). Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence. *Official Journal of the European Union*, L, 2024/1689. <https://eur-lex.europa.eu/eli/reg/2024/1689/oj>
- Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security* (pp. 79-90). <https://doi.org/10.1145/3605764.3623985>
- Hagendorff, T., Dasgupta, I., Binz, M., Chan, S. C. Y., Lampinen, A., Wang, J. X., Akata, Z., & Schulz, E. (2023). Machine psychology. arXiv. <https://doi.org/10.48550/arXiv.2303.13988>
- Ho, M. K., Zhu, Z., Zhu, R., Li, L., Fan, Z., Wang, Z., & Hong, J. (2026). Can induced emotion bias LLM behaviors in sequential decision making? arXiv:2607.12631. <https://doi.org/10.48550/arXiv.2607.12631>
- Kron, A., Goldstein, A., Lee, D. H.-J., Gardhouse, K., & Anderson, A. K. (2013). How are you feeling? Revisiting the quantification of emotional qualia. *Psychological Science*, 24(8), 1503-1511. <https://doi.org/10.1177/0956797613475456>
- Lerer, I. A. (2026a). Static permissibility in dynamic normative environments: Dynamic classification failure as a structural property of institutional computation. Zenodo. <https://doi.org/10.5281/zenodo.21120096>
- Lerer, I. A. (2026b). Sub-propositional affiliation signals as alignment vectors: An extended phenotype account of kinship vocatives, sycophancy, and the limits of content-level AI safety. Zenodo. <https://doi.org/10.5281/zenodo.21166093>
- Liu, Y., Deng, G., Li, Y., Wang, K., Wang, Z., Zhang, T., Liu, Y., Wang, H., Zheng, Y., et al. (2023). Prompt injection attack against LLM-integrated applications. arXiv. <https://doi.org/10.48550/arXiv.2306.05499>
- Maia, T. V., & McClelland, J. L. (2004). A reexamination of the evidence for the somatic marker hypothesis: What participants really know in the Iowa Gambling Task. *Proceedings of the National Academy of Sciences*, 101(45), 16075-16080. <https://doi.org/10.1073/pnas.0406666101>

National Institute of Standards and Technology. (2023). *Artificial Intelligence Risk Management Framework (AI RMF 1.0)* (NIST AI 100-1). <https://doi.org/10.6028/NIST.AI.100-1>

National Institute of Standards and Technology. (2024). *Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile* (NIST AI 600-1). <https://doi.org/10.6028/NIST.AI.600-1>

Posner, J., Russell, J. A., & Peterson, B. S. (2005). The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and Psychopathology*, 17(3), 715-734. <https://doi.org/10.1017/S0954579405050340>

Russell, J. A. (1980). A circumplex model of affect. *Journal of Personality and Social Psychology*, 39(6), 1161-1178. <https://doi.org/10.1037/h0077714>

Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., & Yao, S. (2023). Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*, 36. https://proceedings.neurips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations*. https://openreview.net/forum?id=WE_vluYUL-X